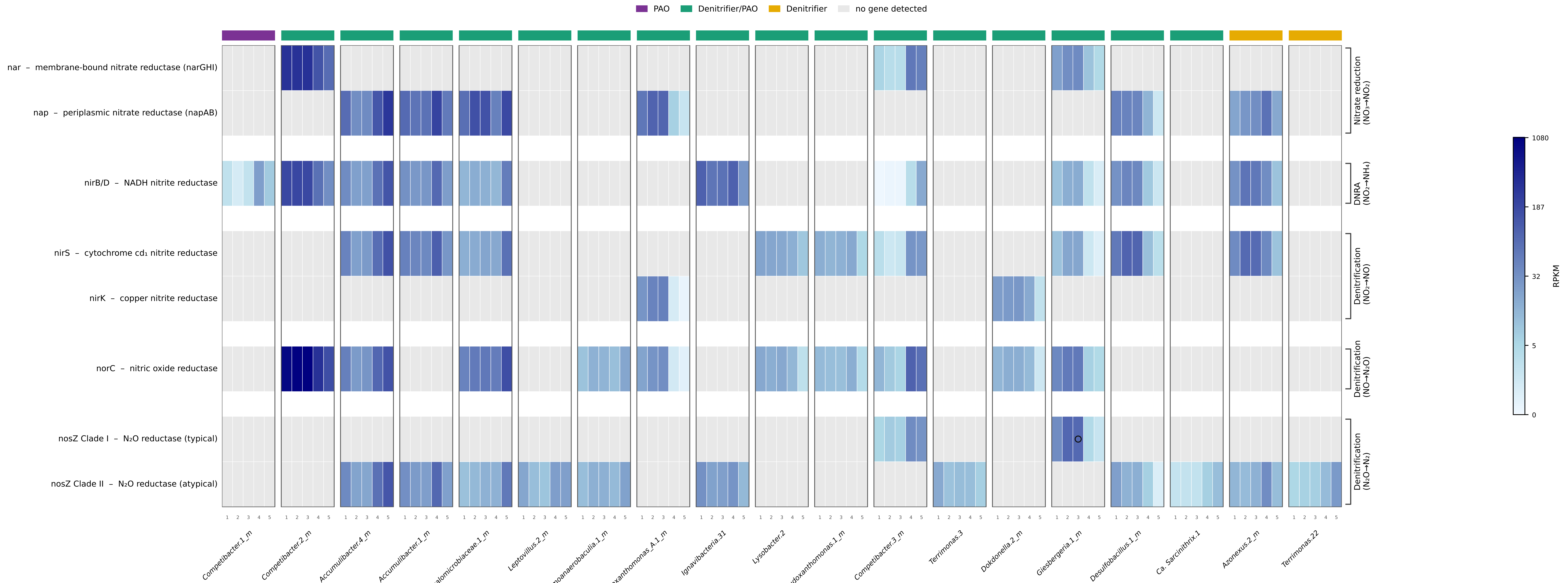
Denitrification & DNRA gene abundance across top high-quality MAGs of the CAN system



Per-sample gene abundances in reads per kilobase per million mapped reads (RPKM) from long-read (Nanopore) metagenomes of the CAN system. Columns are grouped by MAG — the union of the ten most abundant high-quality MAGs per sample, labelled by MAG iterativeID — each subdivided into the five samples CAN_1–CAN_5 (sub-columns 1–5, left→right); the coloured bar marks GAO / PAO / denitrifier classification (mag_abundance_summary.tsv logic). Rows are denitrification/DNRA KO functions grouped by nitrogen-transformation stage (right brackets); for the multi-gene rows — nar (narGHI; K00370/K00371/K00374), nap (napAB; K02567/K02568), nirB/D (K00362/K00363) and norC (norC/norB; K02305/K04561) — the cell is the maximum RPKM over the row's genes. nosZ (K00376) is split into Clade I (typical denitrifier) and Clade II (atypical / non-denitrifier) by per-gene nhmmer classification against the NosZREF Clade I and Clade II (sub-clades 1577 A –H) reference alignments; ○ marks a low-confidence clade call. Grey cells indicate no detected gene.